

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11: GROWTH AND DEVELOPMENT IN PLANTS

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the driving question: What causes the differences in germination, emergence and growth of maize and bean plants? Think back to Mr. Wafula's farm near Kitale. He planted maize H614 and Rosecoco beans on the same day (Day 0), 3 cm deep, after 25 mm of rain. Use the evidence you collected in Lessons 1 to 10. Write your answers in the space under each prompt. Use full sentences, include numbers from the data, and use the words germination, emergence, dormancy, viable, primary growth and hormones correctly. Where a prompt asks for Claim, Evidence and Reasoning, write all three parts. Where a data table is given, use it. Work on your own.

Part 1: Seed structure and emergence (CER)

Think about the time-lapse videos of the bean and maize seedlings that you watched. Write a Claim-Evidence-Reasoning explanation of how the different seed structures of maize H614 and Rosecoco beans lead to different ways of emerging. In your answer: (a) name the type of seed each crop has and how many cotyledons it has; (b) say where each crop stores its food; (c) use what you saw in the time-lapse videos to say what happens to the cotyledon(s) of each crop during germination and what this type of germination is called, and give the day each crop emerged above the soil; (d) explain the difference between germination and emergence.

Claim: Maize and bean seedlings emerge differently because their seeds have different structures. Evidence: Maize is a monocot with one cotyledon (the scutellum). It stores food in the endosperm, and the scutellum absorbs it. The coleoptile protects the shoot, and emergence was on Day 7. The bean is a dicot with two cotyledons that store the food. In the time-lapse, the bean radicle appeared on Day 3 and the bean emerged on Day 8, when an arched hypocotyl lifted the cotyledons above the ground. The maize radicle appeared on Day 4, and the scutellum stayed underground. Reasoning: Germination is when the radicle comes out of the seed. Emergence is when the shoot appears above the soil. They are different events, so a seed can germinate days before it emerges. In maize the cotyledon stays underground, which is hypogeal germination. In beans the hypocotyl arch lifts the cotyledons above ground, which is epigeal germination. The way each seedling reaches the surface comes from its seed structure and not from the rain or planting date, which were the same for both.

Part 2: Why healthy-looking seeds did not germinate

Use the table from Mr. Wafula's field at Day 14 (100 seeds of each crop).

Reason seed did not germinate	Maize H614	Bean Rosecoco
Eaten or rotted by fungi/pests	5	2
Old seed with dead embryo (not viable)	4	4
Waterlogged low patch (no oxygen)	3	6
Hard impermeable seed coat	0	8
Total that did not germinate	12	20

Claim: The seeds failed for different reasons, so dormancy explains only some of the healthy-looking seeds that did not germinate. Evidence: Only the 8 beans with a hard impermeable seed coat were dormant, and 0 maize seeds were dormant. These 8 were counted as not germinated at Day 14. In the later re-test, after they were scarified (nicked) and replanted, 7 of the 8 germinated by Day 7. The other failures were 5 maize and 2 beans eaten or rotted by fungi or pests, 4 maize and 4 beans with dead embryos, and 3 maize and 6 beans in the waterlogged low patch. Check: $5+4+3 = 12$ maize and $8+6+4+2 = 20$ beans. Reasoning: Dormancy means a viable (living) seed fails to germinate even in favourable conditions. The 8 beans were viable, and breaking the testa let them take in water, so the hard seed coat was the cause. The seeds with dead embryos were not viable, so they could never germinate. Seeds eaten by pests or rotted were destroyed. Seeds in the waterlogged patch were not dormant; their failure was linked to the lack of oxygen in that patch. Absciscic acid maintains dormancy, and gibberellins promote germination. So seed condition and environment, not rain or planting date, explain why healthy-looking seeds did not germinate.

Write a CER explanation of why 12 maize and 20 bean seeds did not germinate. In your answer: (a) say which seeds were dormant and what dormancy means; (b) the 8 hard-coated bean seeds were later scarified and replanted as a separate re-test after Day 14. Say what happened to them and what this shows about the cause of their dormancy; (c) explain why the other failed seeds were not dormant; (d) name the hormone that maintains dormancy and the hormone that helps germination.

Part 3: Conditions needed for germination

Fatuma set up germination jars with 20 Rosecoco bean seeds in each jar. These are the Day 5 results.

Jar	Condi ons	Seeds germin ated (out of 20)
A	Water + air + 25 C	18
B	Dry	0
C	No air (subme rged in boiled, cooled water	1

Claim: A viable, non-dormant seed needs water, oxygen (air) and a suitable temperature to germinate. Evidence: In Jar A, which had water, air and 25 C, 18 of 20 seeds germinated. When one condition was missing, almost none germinated: 0 in the dry jar B, 1 in jar C with no air, and 0 in the cold jar D at 4 C. Reasoning: Water activates the enzymes and metabolism in the embryo, and the stored food then supplies energy and materials for the growing radicle. Oxygen is needed for respiration to release that energy. A suitable temperature lets the enzymes work. If any one is missing, germination fails. In Mr. Wafula's waterlogged low patch, 3 maize and 6 bean seeds failed. Waterlogged soil has little oxygen, so the lack of oxygen is the likely reason, as jar C suggests, although we did not test the patch itself. The jar and the patch are alike because in both, a lack of oxygen is linked to seeds failing to germinate. They are different because the jar was a controlled test with 20 seeds and one condition changed at a time, while the patch was part of a field where we only counted the 3 maize and 6 bean seeds that failed and did not measure the oxygen. This shows that conditions in a particular spot in the field matter, not just the date or the rainfall.

	with oil layer)	
D	Cold (fridge, 4 C)	0

Write a CER explanation of which conditions a viable, non-dormant seed needs to germinate. Then use the table, and the numbers of seeds that failed in the waterlogged low patch of Mr. Wafula's field, to explain what may have happened there. Say how the jar and the field are alike and how they are different.

Part 4: Growth after emergence

After emergence the two crops grew differently. Use this data from Mr. Wafula's field (mean shoot height of 10 seedlings, in cm).

D a y	7	14	21	28	35	42
M a i z e	3	12	30	62	98	125
B	4	9	1	3	4	4

Claim: Maize grows faster than bean, and its growth was still rapid at Day 42, while bean growth had slowed by then. Both follow the sigmoid pattern of growth, but the data show different parts of it. Evidence: Maize was taller from Day 14 onward and reached 125 cm by Day 42, while bean reached 44 cm. Maize grew most between Day 28 and Day 35 (36 cm, from 62 to 98 cm), while bean grew most between Day 21 and Day 28 (12 cm, from 18 to 30 cm). Bean growth then slowed to 4 cm between Day 35 and Day 42. Maize still grew 27 cm between Day 35 and Day 42, which is less than the 36 cm before. Stem width rose from 6 to 22 mm in maize and from 2 to 6 mm in bean. Reasoning: A sigmoid growth curve has a lag phase, a log (rapid, exponential) phase and a stationary phase. Maize has its steepest growth between about Day 21 and Day 35, and bean between about Day 14 and Day 28 before flattening, so the bean data show a slowing towards the stationary phase. Maize is only starting to slow by Day 42 and still rises by 27 cm in the last week, so we cannot yet see a stationary phase for maize in this data. Primary growth is growth in length of roots and shoots. It happens at the apical meristems, where cells divide, then elongate, then differentiate. Maize is a monocot with scattered vascular bundles and no vascular cambium, so it has no true secondary growth and makes no wood. Its stem thickens through primary growth. Secondary growth by the vascular cambium happens in woody dicots such as *Grevillea robusta*, which show annual rings.

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Stem width at Day 14, 28 and 42 was 6, 14 and 22 mm for maize and 2, 4 and 6 mm for bean. Write a CER explanation of how the growth of maize and bean differ. In your answer: (a) say between which days each crop grew the most and how many centimetres it grew in that period; (b) say what shape a growth curve has and name the three phases, then say which phases you can see in the bean data and in the maize data up to Day 42; (c) explain what primary growth is and where it happens; (d) explain why maize gets thicker even though it does not make wood.

Part 5: Hormones, apical dominance and your final model

Mr. Wafula compared two groups of Rosecoco bean plants. In one group the shoot tip was left intact; in the other the shoot tip was removed at Day 20. Mr. Wafula counted the branches on each group at Day 25 and Day 32. (a) Using what

(a) Claim: The bean branched only when the tip was removed because the shoot tip stops the lateral buds from growing. Evidence: With the tip intact there was 1 main stem and 0 branches at Day 25. When the tip was removed at Day 20, 3 lateral branches grew by Day 32. Reasoning: The apical meristem makes auxin (IAA), which moves down the stem and inhibits the lateral buds. This is apical dominance. Removing the tip lowers the auxin, so cytokinins can promote the growth of lateral buds into branches. (b) Auxin is made in the shoot tip and inhibits the lateral buds, and cytokinins promote lateral buds. (c) The differences are not caused by the rain or planting date, which were the same for both crops. They come from seed structure (maize is a monocot with hypogeal germination and emerged on Day 7; bean is a dicot with epigeal germination and emerged on Day 8), seed condition (8 dormant bean seeds with a hard seed coat, plus separate failures from non-viable seeds with dead embryos and from pest damage) and

you learned about hormones, write a CER explanation of what you would expect the two groups to look like and why. Use the numbers of branches from Lesson 7. (b) Name the hormone made in the shoot tip that affects lateral buds, say what it does to them, and name the hormone that promotes lateral buds. (c) Now write a short final answer to the driving question: What causes the differences in germination, emergence and growth of maize and bean plants? Finish by saying two ways in which your final model is different from your Day 1 drawing.	the environment (water, oxygen and temperature, and the waterlogged patch). After germination, primary growth at meristems lengthens roots and shoots, and hormones such as auxin, gibberellins, cytokinins and abscisic acid coordinate growth and development. Growth is not only getting taller: plants also thicken, branch and change in form. My Day 1 model only said the differences might be due to the seed, soil, water, age or pests. My final model is different because I added dormancy and the hard seed coat, the three conditions needed for germination, the growth curve, and hormones such as auxin controlling branching.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy	All terms (germination, emergence, dormancy, viable, primary growth, hypogeal, epigeal, hormones) are used correctly, with no errors, and the maize/bean differences are right.	Most terms and ideas are correct, with one or two small errors that do not change the meaning.	Several ideas are wrong or terms are mixed up, for example confusing germination with emergence or saying maize makes wood.
Use of evidence	Uses exact data from Mr. Wafula's farm in every section, such as 88 and 80 germinated, 8 dormant beans with 7 germinating, the jar results, heights and branches.	Uses some data from the unit, but misses a part or quotes a number loosely.	Gives few or no numbers from the unit, or the numbers do not match the data.
Reasoning	Clearly links each piece of evidence to a biological idea, explaining why it supports the claim, such as water and enzymes, oxygen and respiration, and auxin and lateral buds.	Links most evidence to ideas, but some links are brief or incomplete.	Lists facts without explaining how they support the claim.
Connection to the driving question and model revision	Gives a complete answer to the driving question covering seed structure, seed condition, environment and hormones, and says two clear changes from the Day	Answers the driving question with most causes and gives at least one change from the Day 1 model.	The answer covers only one cause or does not say how the model changed.

	1 model.		
Clarity of communication	Well organised Claim, Evidence and Reasoning in clear sentences, with data read correctly from the tables.	Mostly clear and organised, with a few unclear sentences.	Hard to follow, with the Claim, Evidence or Reasoning missing.

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